Axis	**PSR**	**NSR**	**FD (baseline)**
generator	$A_j^2=I$ (Pauli)	$\mathrm{diam}(A_j)<\infty$	none
hardware	op + gate zone/branch	headroom $s_0=\pi/2\Omega$	none
fine params ($\Delta t\to 0$)	✓	× ($s_0\propto 1/\Delta t$)	✓
executions / grad.	$O(m)$ pairs	$O(M)$ / $O(1)^s$	$O(1)$
shots ($/\varepsilon_g^2 e^{2\Gamma T}$)	$O(D_1^2)$	$O(\chi^2 D_1^2)$	$O(1/\varepsilon^2)$, m-indep.
compilation	waveforms + gates	waveform only	waveform only
bias	0; gate $\le C_{\mathrm{PSR}}\varepsilon_{\mathrm{ins}}$	$O(\Omega/M)$ / 0^s	$O(\varepsilon^2)+O(\delta/\varepsilon)$
unbiased, noisy ∇C	✓	✓†	×
coherent-error suppr.	✓† $O(\eta^2)$	†	× $O(\eta)$

$\dot{H}=\sum_j(\partial u_j/\partial\theta)A_j$; $D_1=\int_0^T\sum_j|\partial u_j/\partial\theta|\mathrm{diam}(A_j)dt$; $\Omega=\int_0^T\mathrm{diam}(\dot{H})dt$; $\chi=\Omega/D_1\in(0,1]$; T time; $\Gamma=1/T_2$; m ext.-θ terms; M trunc. order; ε_g target RMSE; ε FD step; δ control-amp. error; η coherent-error; $\varepsilon_{\mathrm{ins}}$ gate infidelity; $s_{\max}$ headroom; † numeric (proof open); ^{s} stochastic sampler.